

# Quiz — Power and Efficiency

Physics · Grade 11 · Lesson 1–6 · Multiple choice · 5 questions

Mr. Gemy  
Time: 15 min  
Total: 5 marks

Name: .....

Class: .....

Date: .....

**Data:**  $g = 9.8 \text{ m/s}^2$  |  $\cos 37^\circ = 0.8$  |  $\cos 60^\circ = 0.5$ .

**Instructions:** choose the **one** correct answer for each question and write its letter in the box.

**Q1** Lift A raises a load of **400 kg** through **15 m** in **30 s**. Lift B raises the **same load** through the **same height** in **20 s**. Which statement is correct?

- ☐ A) lift B does 1.5 times the work done by lift A
- ☐ B) they do the same work, and the power of B is 1.5 times the power of A
- ☐ C) they have the same power, and lift B does more work
- ☐ D) they do the same work, and the power of A is 1.5 times the power of B

**Q2** A rope pulls a box along a horizontal floor at a constant speed of **5 m/s** with a force of **60 N** that makes  **$37^\circ$**  with the direction of motion (Fig. 1). The power of the force is:

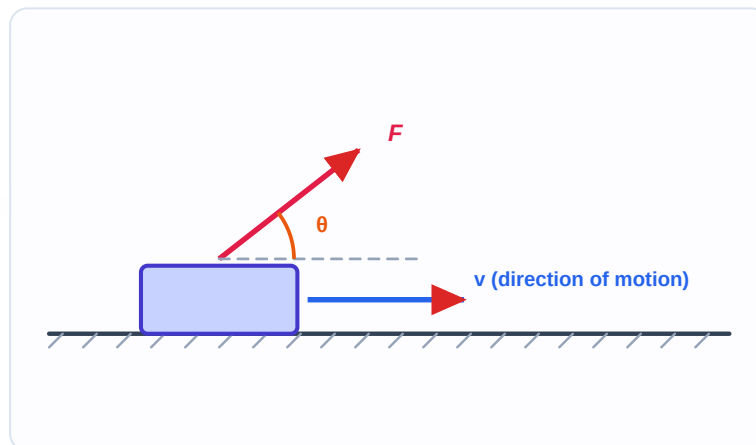


Fig. 1

- |                                |                                |
|--------------------------------|--------------------------------|
| <input type="radio"/> A) 300 W | <input type="radio"/> B) 180 W |
| <input type="radio"/> C) 240 W | <input type="radio"/> D) 12 W  |

**Q3** An electric kettle takes a power of **2500 W** and gives the water a useful thermal power of **2000 W**. Its efficiency is:

- |                                |                               |
|--------------------------------|-------------------------------|
| <input type="radio"/> A) 125 % | <input type="radio"/> B) 80 % |
| <input type="radio"/> C) 20 %  | <input type="radio"/> D) 50 % |

**Q4** A motor of efficiency **70 %** raises a load of **140 kg** at a **constant speed** of **0.25 m/s** (Fig. 2). The electric power taken by the motor is:

★ challenging

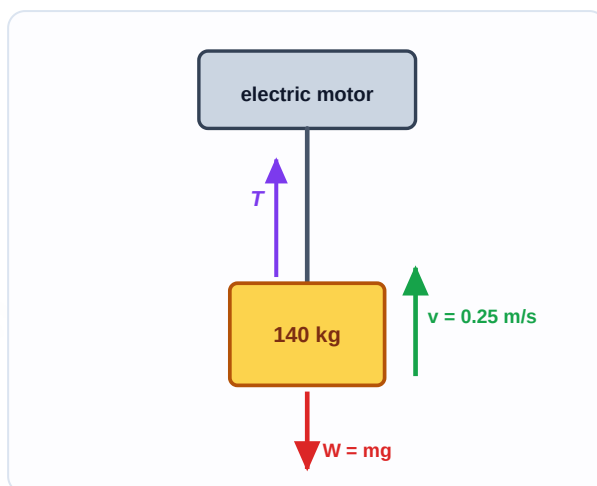


Fig. 2

A) 343 W

B) 490 W

C) 240.1 W

D) 1372 W

**Q5** Fig. 3 shows the useful work done by two machines X and Y against time. The efficiency of X is **80 %** and the efficiency of Y is **60 %**. The ratio of the **input** powers of the two machines (  $P_{in}(X) / P_{in}(Y)$  ) is:

★ challenging

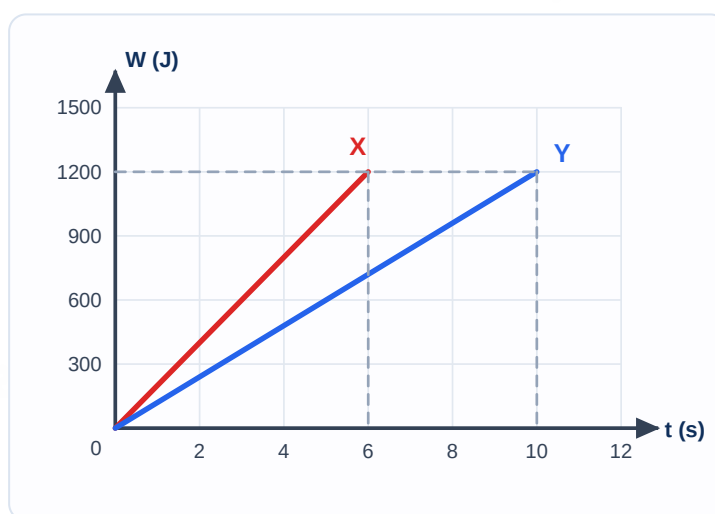


Fig. 3

A) 1.25

B) 5/3

C) 0.8

D) 2.22